Code No: 134BU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, September/October - 2023
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) What is a system software? How is it different from an application software? [2]
b) Draw the layered architecture of an operating system and explain various components of it. [3]
c) What is meant by context switch? [2]
d) Distinguish between ordinary pipes and named pipes. [3]
e) What is demand paging? [2]
f) Compare LRU page replacement algorithm with optimal page replacement algorithm. [3]
g) What are magnetic disks? [2]
h) List and explain various file open access modes. [3]
i) What are the goals of protection? [2]
j) What are capability based systems? [3]

PART – B

(50 Marks)

2. Explain different types of operating system structures. [10]

OR

3. What is a system call? What are different types of system calls? Explain them. [10]

4.a) Define the term scheduler. Explain briefly about long term, medium term, and short term schedulers with proper diagram.
b) What is process control block? Explain various fields in it. [5+5]

OR

5. How to implement synchronization between the processes using semaphores? Explain with an illustrative example. [10]

6.a) What is meant by thrashing? Explain various methods to overcome thrashing.
b) Explain various frame allocation methods using examples. [5+5]

OR

JNTUH USED PAPERS 2023

7. Using FIFO, LRU and optimal page replacement algorithms on following reference string.
1,2,3,2,1,2,3,4,5,3,6,7,4,3,5.
Compute the number of page faults for the frame size of 3 and 4. [10]

8. Explain FCFS and SSTF disk scheduling algorithms using examples. [10]

OR

9. Explain about file system mounting and file sharing. [10]

10. Explain banker's algorithm for deadlock avoidance with an example. [10]

OR

11. Explain the implementation of access matrix in detail. [10]

---ooOoo---

JNTUH USED PAPERS 2023